Table 1: When does a learned KKL observer win (C3)? id, dev plants: number of plants on which HyperKKL ($r = 4$, the single learned method) beats the comparator (paired 95% CI excludes 0). Per-plant numbers: Table **??**.

| | | (a) vs model-based filters | | | | | (b) vs data-driven | |
|---|---|---|---|---|---|---|---|---|
| scenario | model err. | best filter | working | nominal | rob./orac. | failed | best data-driven | NLOX |
| exact | 0.00 | 0/6 (6L) | 0/6 (6L) | 0/6 (6L) | 0/6 (6L) | 1/48 | pend. 0/30 | 3/6 |
| exact+outliers | 0.00 | 0/6 (6L) | 0/6 (6L) | 0/6 (6L) | 0/6 (6L) | 13/48 | pend. 0/30 | 1/6 (4L) |
| colored_t | 0.00 | 0/6 (6L) | 0/6 (6L) | 0/6 (6L) | 0/6 (6L) | 1/48 | 0/6 (6L) | 4/6 |
| mismatch10 | 0.09 | 5/6 (1L) | 5/6 (1L) | 5/6 (1L) | 5/6 (1L) | 13/48 | 0/6 (4L) | 3/6 |
| mismatch20 | 0.17 | 5/6 (1L) | 4/5 (1L)* | 5/6 (1L) | 5/6 (1L) | 15/48 | 1/6 (4L) | 4/6 |
| drift30 (post-train) | 0.24 | 0/6 (5L) | 0/5 (5L)* | 1/6 (3L) | 1/6 (5L) | 15/48 | pend. 0/30 | 2/6 (1L) |
| mismatch30 | 0.24 | 5/6 (1L) | 4/5 (1L)* | 5/6 (1L) | 5/6 (1L) | 15/48 | 0/6 (4L) | 3/6 |
| mismatch50 | 0.37 | 6/6 | 4/4* | 6/6 | 6/6 | 23/48 | 0/6 (4L) | 3/6 |
| partial | 0.44 | 5/6 (1L) | 4/5 (1L)* | 5/6 (1L) | 5/6 (1L) | 14/48 | 0/6 (4L) | 3/6 (1L) |
| realistic | 0.63 | 6/6 | 4/4* | 6/6 | 6/6 | 24/48 | 1/6 (4L) | 5/6 (1L) |

($k$L): the comparator is significantly better on $k$ plants. (a) Model-based filters: EKF, UKF, PF, KKL-MB and their Huber and oracle variants. A filter is FAILED when its NRMSE exceeds climatology (the constant mean-state predictor); 'working' = the best filter that did not fail (*: no filter works on some plants). (b) Data-driven baselines trained on the same real-plant data: NLOX, Koopman KF (incl. bilinear), grey-box EKF/UKF/EKF-oracle; 'pend. $m/N$' = merged in $m$ of $N$ runs. Model error: median over dev plants.